

20251027 Numerical Bathtub

1. Model build-up

Let's consider solving the time-dependent regulator model numerically. We start with writing down the set of ODEs and note that I replace all the parameters into surface density to describe the spatially-resolved scenario.

First, as usual, we define the SFR surface density in two ways:

$$\Sigma_{\text{SFR}}[t] = \epsilon[t]\Sigma_g[t] = \frac{d}{dt}\Sigma_*[t] \quad (1)$$

Then, the change of gas mass surface density:

$$\frac{d}{dt}\Sigma_g[t] = \Sigma_{\text{in}}[t] - (1 - R + \lambda)\epsilon[t]\Sigma_g[t] \quad (2)$$

Next, the change of metal mass surface density:

$$\frac{d}{dt}\Sigma_Z[t] = y(1 - R)\epsilon[t]\Sigma_g[t] - (1 - R + \lambda)\epsilon[t]\Sigma_g[t]Z[t] + \Sigma_{\text{in}}[t]Z_{\text{in}} \quad (3)$$

And, we also define the metallicity as:

$$Z[t] = \frac{\Sigma_Z[t]}{\Sigma_g[t]} \quad (4)$$

Also, the sSFR:

$$\text{sSFR} = \frac{\Sigma_{\text{SFR}}}{\Sigma_*} \quad (5)$$

Finally, we can convert metallicity to observable with solar abundace at 8.69 and $Z_{\text{surface}} = 0.0139$ (Asplund+2021):

$$12 + \log_{10}(\text{O/H})[t] = 8.69 + \log_{10}\left(\frac{Z[t]}{0.0139}\right) \quad (6)$$

2. Equilibrium state

Taking the product rule of Eq. 3 and substituting Eq.1 to it, we get $\frac{dZ[t]}{dt}$

$$\begin{aligned} Z[t](\Sigma_{\text{in}}[t] - (1 - R + \lambda)\Sigma_{\text{SFR}}[t]) + \Sigma_g[t]\frac{dZ[t]}{dt} &= y(1 - R)\Sigma_{\text{SFR}}[t] - Z[t](1 - R + \lambda)\Sigma_{\text{SFR}}[t] + \\ \Sigma_g[t]\frac{dZ[t]}{dt} &= y(1 - R)\Sigma_{\text{SFR}}[t] + (Z_{\text{in}} - Z[t])\Sigma_{\text{in}}[t] \\ \frac{dZ[t]}{dt} &= \frac{1}{\Sigma_g[t]}(y(1 - R)\Sigma_{\text{SFR}}[t] + (Z_{\text{in}} - Z[t])\Sigma_{\text{in}}[t]) \end{aligned}$$

Now let's consider the equilibrium state that the model rapidly self-regulates to a steady state, which means $\frac{dZ[t]}{dt} = 0$ in a short time scale (compared to the total depletion time). Thus, such metallicity at equilibrium state (Z_{eq}) are a series of constant that is time-independent in a short period of time (but it is still time-dependent in a much longer period, so it is still written as $Z_{\text{eq}}[t]$). And therefore it requires

$$\begin{aligned}\frac{dZ[t]}{dt} &= \frac{1}{\Sigma_g[t]} (y(1-R)\Sigma_{\text{SFR}}[t] + (Z_{\text{in}} - Z_{\text{eq}}[t])\Sigma_{\text{in}}[t]) = 0 \\ y(1-R)\Sigma_{\text{SFR}}[t] + (Z_{\text{in}} - Z_{\text{eq}}[t])\Sigma_{\text{in}}[t] &= 0 \\ Z_{\text{eq}}[t] &= Z_{\text{in}} + y(1-R)\frac{\Sigma_{\text{SFR}}[t]}{\Sigma_{\text{in}}[t]}\end{aligned}\quad (8)$$

Again, we are interested in the metallicity that satisfy $Z[t] = Z_{\text{in}} + y(1-R)\frac{\Sigma_{\text{SFR}}[t]}{\Sigma_{\text{in}}[t]}$, at any given t .

Thus, with this set of Z_{eq} , we can find the mean value ($\langle Z_{\text{eq}} \rangle$) and then consider a small perturbation (offset from $\langle Z_{\text{eq}} \rangle$). Since y and R are all constant here, we have

$$\delta Z_{\text{eq}}[t] = y(1-R)\delta\left(\frac{\Sigma_{\text{SFR}}[t]}{\Sigma_{\text{in}}[t]}\right) \quad (9)$$

Because we are interested in the offset in $12 + \log_{10}(\text{O}/\text{H})$, we have

$$\begin{aligned}\delta(12 + \log_{10}(\text{O}/\text{H})_{\text{eq}}[t]) &= \delta \log_{10}(Z_{\text{eq}}[t]) \\ &= \frac{1}{\ln 10} \frac{\delta Z_{\text{eq}}[t]}{Z_{\text{eq}}[t]} \\ &= \frac{1}{\ln 10} \frac{y(1-R)\delta\left(\frac{\Sigma_{\text{SFR}}[t]}{\Sigma_{\text{in}}[t]}\right)}{Z_{\text{in}} + y(1-R)\frac{\Sigma_{\text{SFR}}[t]}{\Sigma_{\text{in}}[t]}} \\ &= \frac{y(1-R)\frac{\Sigma_{\text{SFR}}[t]}{\Sigma_{\text{in}}[t]}}{Z_{\text{in}} + y(1-R)\frac{\Sigma_{\text{SFR}}[t]}{\Sigma_{\text{in}}[t]}} (\delta \log_{10}(\Sigma_{\text{SFR}}[t]) - \delta \log_{10}(\Sigma_{\text{in}}[t])) \\ &= \frac{Z_{\text{eq}}[t] - Z_{\text{in}}}{Z_{\text{eq}}[t]} (\delta \log_{10}(\Sigma_{\text{SFR}}[t]) - \delta \log_{10}(\Sigma_{\text{in}}[t])) \\ &\propto \delta \log_{10}(\Sigma_{\text{SFR}}[t]) - \delta \log_{10}(\Sigma_{\text{in}}[t])\end{aligned}\quad (10)$$

Thus, we can see that the sign of $\delta(12 + \log_{10}(\text{O}/\text{H})_{\text{eq}})$ is only determined by relative change of $\log_{10}(\Sigma_{\text{SFR}}[t])$ and $\log_{10}(\Sigma_{\text{in}}[t])$.

More specific, by $\Sigma_{\text{SFR}}[t] = \epsilon[t]\Sigma_g[t]$, where $\epsilon[t]$ is the star formation efficiency, we have

$$\delta(12 + \log_{10}(\text{O}/\text{H})_{\text{eq}}) \propto \delta \log_{10}(\epsilon[t]) + \delta \log_{10}(\Sigma_g[t]) - \delta \log_{10}(\Sigma_{\text{in}}[t]) \quad (11)$$

Therefore, this leaves us 3 cases. Note that δ in log scale of a quantity indicates the relative change of the quantity, and the sign indicates increase or decrease.

1. In the case that $\delta \log_{10}(\epsilon[t])$ dominants, if $\delta \log_{10}(\epsilon[t])$ increases, both $\delta(12 + \log_{10}(\text{O}/\text{H})_{\text{eq}})$ and $\delta \log_{10}(\Sigma_{\text{SFR}}[t])$ increase immediately, so we can observe the positive correlation between them.
2. However, if $\delta \log_{10}(\Sigma_{\text{in}}[t])$ dominates, even though $\delta \log_{10}(\Sigma_{\text{SFR}}[t])$ is still positive, due to the minus sign and relatively large change in $\delta \log_{10}(\Sigma_{\text{in}}[t])$, we can observe negative correlation.
3. If the change of $\log_{10}(\Sigma_{\text{SFR}}[t])$ and $\log_{10}(\Sigma_{\text{in}}[t])$ is comparable, that lead to

nearly stationary $12 + \log_{10}(\text{O}/\text{H})$, which show no correlation. In this scenario, it requires $\Phi[t]$ is linearly proportional to $\text{SFR}[t]$.

3. Constants and initial values

Here, return fraction, metal yield per stellar, mass loading factor and inflow metallicity are constants:

$$\begin{cases} R = 0.5 \\ y = 0.01 \\ \lambda = 1.5 \\ Z_{\text{in}} = 0.1 \end{cases} \quad (12)$$

Now let's say at $t = 0\text{Myr}$, we have

$$\left\{ \begin{array}{l} \epsilon_0 = \epsilon[t = 0] = \frac{\Sigma_{\text{SFR},0}}{\Sigma_{\text{g},0}} = 10^{-9}/\text{yr} \\ \Sigma_{\text{in},0} = \Sigma_{\text{in}}[t = 0] = 10^{-3}\text{M}_{\odot}/\text{yr}/\text{kpc}^2 \\ Z_0 = Z[t = 0] = 0.5 \\ \Sigma_{*,0} = \Sigma_{*}[t = 0] = 10^7\text{M}_{\odot}/\text{kpc}^2 \\ \Sigma_{\text{g},0} = \Sigma_{\text{g}}[t = 0] = 10^6\text{M}_{\odot}/\text{kpc}^2 \\ \Sigma_{\text{SFR},0} = \Sigma_{\text{SFR}}[t = 0] = 10^{-3}\text{M}_{\odot}/\text{yr}/\text{kpc}^2 \\ \text{sSFR}_0 = \text{sSFR}[t = 0] = \frac{\Sigma_{\text{SFR},0}}{\Sigma_{*,0}} = 10^{-10}/\text{yr} \\ \Sigma_{Z,0} = \Sigma_Z[t = 0] = \Sigma_{\text{in},0}\Sigma_{Z,0} = 0.5 \times 10^6\text{M}_{\odot}/\text{kpc}^2 \\ 12 + \log_{10}(\text{O}/\text{H})_0 = 12 + \log_{10}(\text{O}/\text{H})[t = 0] = 8.69 + \log_{10}\left(\frac{Z_0}{0.0139}\right) \end{array} \right. \quad (13)$$

4. Numerical forms of $\epsilon[t]$ and $\Sigma_{\text{in}}[t]$

We consider the following cases:

4.1 Constant $\epsilon[t]$ and constant $\Sigma_{\text{in}}[t]$

A simple case that both are constant:

$$\begin{cases} \epsilon[t] = \epsilon_0 = 10^{-9}/\text{yr} \\ \Sigma_{\text{in}}[t] = \Sigma_{\text{in},0} = 10^{-3}\text{M}_{\odot}/\text{yr}/\text{kpc}^2 \end{cases} \quad (14)$$

4.2 Sinusoidal variation of $\epsilon[t]$ and constant $\Sigma_{\text{in}}[t]$

Here we define a constant $A = 0.1$ here to denote the amplitude of the sinusoidal variation compared to initial value, and $T_p = 1000$ (in the unit of Myr) to indicate the variation period:

$$\begin{cases} \epsilon[t] = \epsilon_0 + A\epsilon_0 \sin\left(\frac{2\pi t}{T_p}\right) \\ \Sigma_{\text{in}}[t] = \Sigma_{\text{in},0} = 10^{-3}\text{M}_{\odot}/\text{yr}/\text{kpc}^2 \end{cases} \quad (15)$$

4.3 Constant $\epsilon[t]$ and sinusoidal variation of $\Sigma_{\text{in}}[t]$

Similarly, we have:

$$\begin{cases} \epsilon[t] = \epsilon_0 = 10^{-9}/\text{yr} \\ \Sigma_{\text{in}}[t] = \Sigma_{\text{in},0} + A\Sigma_{\text{in},0} \sin\left(\frac{2\pi t}{T_p}\right) \end{cases} \quad (16)$$

5. Evolution

We consider a total period of $T = 10\text{Gyr}$, with a timestep of $\Delta t = 0.1\text{Myr}$. Thus, for any given moment $t_i \in [0, T]$, the iteration in the next timestamp ($t_{i+1} = t_i + \Delta t$) will be

$$\left\{ \begin{array}{l} \epsilon[t_{i+1}] = \epsilon[t_i] \\ \Sigma_{\text{in}}[t_{i+1}] = \Sigma_{\text{in}}[t_i] \\ \Sigma_{\text{g}}[t_{i+1}] = \Sigma_{\text{g}}[t_i] + \frac{d}{dt}\Sigma_{\text{g}}[t_i]\Delta t \\ \quad = \Sigma_{\text{g}}[t_i] + (\Sigma_{\text{in}}[t_i] - (1 - R + \lambda)\epsilon[t_i]\Sigma_{\text{g}}[t_i])\Delta t \\ \Sigma_{\text{Z}}[t_{i+1}] = \Sigma_{\text{Z}}[t_i] + \frac{d}{dt}\Sigma_{\text{Z}}[t_i]\Delta t \\ \quad = \Sigma_{\text{Z}}[t_i] + (y(1 - R)\epsilon[t_i]\Sigma_{\text{g}}[t_i] - (1 - R + \lambda)\epsilon[t_i]\Sigma_{\text{g}}[t_i]Z[t_i] + \Sigma_{\text{in}}[t_i])\Delta t \\ \Sigma_{\text{SFR}}[t_{i+1}] = \epsilon[t_{i+1}]\Sigma_{\text{g}}[t_{i+1}] \\ Z[t_{i+1}] = \frac{\Sigma_{\text{Z}}[t_{i+1}]}{\Sigma_{\text{g}}[t_{i+1}]} \\ \Sigma_{*}[t_{i+1}] = \Sigma_{*}[t_i] + \Sigma_{\text{SFR}}[t_i]\Delta t \\ \text{sSFR}[t_{i+1}] = \frac{\Sigma_{\text{SFR}}[t_{i+1}]}{\Sigma_{*}[t_{i+1}]} \\ 12 + \log_{10}(\text{O/H})[t_{i+1}] = 8.69 + \log_{10}\left(\frac{Z[t_{i+1}]}{0.0139}\right) \end{array} \right.$$